BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 290 · Specification gap and misconception check

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a salt add to password storage?

Q2. What is an INNER JOIN result based on?

Q3. Which OS mechanism translates pages to frames?

Q4. Full adder A=1,B=0,carry-in=0: sum and carry, and binary total.

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I Do · Read new knowledge

Key vocabulary

misconception — Incorrect mental model of a mechanism.

counterexample — Case disproving a general claim.

transfer — Apply understanding to new case.

diagnostic — Detect specific gaps in knowledge.

New learning

Understand — When correcting an error, identify the incorrect assumption precisely; a bare True/False is weak diagnostic evidence.

Trace — Connect each correction to a worked example: HTTPS versus phishing, paging versus remote files, key joins versus unmatched rows.

Evaluate — Show transfer by applying the rule to a different problem without being prompted by the original misconception.

Worked example

Correct 'a page fault is a missing website page'.

1. Predict first — A page fault is a memory-management event where required virtual page data is not currently resident.

2. Show method — HTTP 404 is a web server status about a requested resource; the word 'page' refers to different concepts.

We Do · Practise with guidance

Correct 'RLE always compresses text'.

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ABCDEF has six runs of length one:
(A,1)(B,1)(C,1)(D,1)(E,1)(F,1)
```

Predict first

The input has no repeated runs longer than one symbol.

Show method

Encoding value/count pairs may consume more bits than the original; compression success depends on data redundancy.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Correct four statements: 'DNS sends HTML'; 'ROM is volatile'; 'SHA is encryption'; 'INNER JOIN keeps all rows'. [8]

Question 2

Explain why a school's cloud files and its RAM paging system should not be described by one mechanism. [6]

Question 3

Choose three of the previous misconceptions and construct a counterexample for each. [6]

Exit Ticket · Answer before checking

An accurate definition alone guarantees that you can apply a concept.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A distinct random per-account value, improving resilience of password hashing.

Q2. Matching records according to a join condition.

Q3. Page table.

Q4. $1+0+0=1=01_2$; sum 1, carry 0.

You Do pages · Independent questions

Q1. DNS resolves names to addresses [2]; ROM non-volatile [2]; hash is one-way digest [2]; INNER JOIN returns matched rows [2].

Q2. Cloud storage = remote files over network [3]; virtual memory = OS logical address mapping to frames/backing store [3].

Q3. Choose any three of the four misconceptions in Question 1. Award for each: a relevant, concrete counterexample [1] and an accurate mechanism-based explanation [1], maximum 6. Examples: (1) DNS resolves school.example to an IP address, while the web server sends HTML in an HTTP response; (2) UEFI firmware in non-volatile ROM/flash remains stored when the computer is powered off; (3) comparing a SHA-256 digest of a downloaded file with a trusted digest can detect changes, but no decryption of the digest recovers the file; (4) an INNER JOIN of Student and Score excludes a pupil with no matching Score row. Accept other correct examples.

Exit Ticket · Model answer

False. Apply it to an unseen example.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding